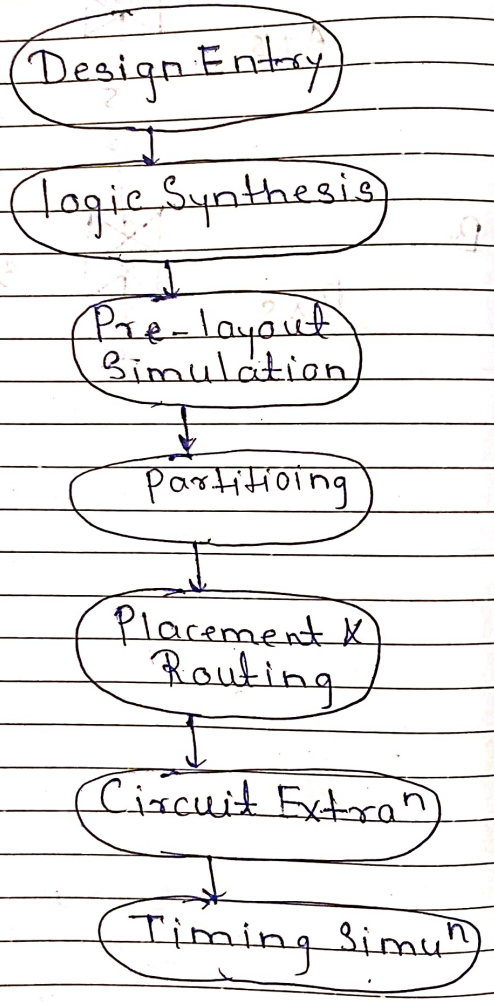
Unit - 5

ASIC design flow



① Design Entry

- we have to entee the design specification, you may use System Schem-atic entey or HDL.

② logic Synthesis

- It Synthesize the design description into the net list.
- means we have to Transform a high-level description into Simple logical blocks.

③ Pre-layout Simulation.

- It is used for logical Corrections.
- It consists Simulation Tool used making Corrections.

④ Partioning

- The entiee System divided into Several Smaller block & floor planning is done.

⑤ Placement & Routing

- This steps decides the location of Cells in a block, as well as making Connection between cell and block

6. Circuit Extraction

- determine the parasitic capacitance as well as resistance.

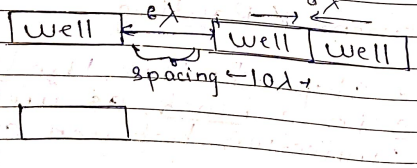
7. Timing Simulation

- check the functionality of a circuit and determine timing analysis.

Lamda Rule

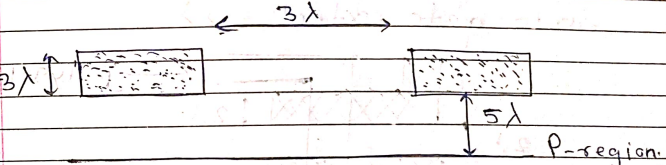
- In the case of MOS-design, different layer are used. many metal connection are present.
- Different contacts are made. These contacts are used for the interconnection bet^n different Thing.
- So certain λ rules are used while designing particular circuit.

① Rule for N-well



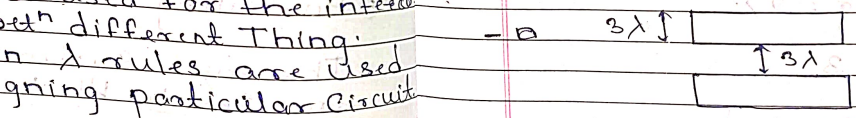
- min width = 10λ.
- cells at same Pote^n with spacing = 6λ
- wells at same poten^n = 0λ
- wells at diff spacing = 8λ

② For active area



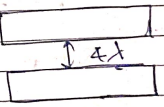
- min width = 3λ.
- min Spacing = 3λ
- well edge spacing = 5λ.

③ Rules For Metal 1



- min width = 3λ
- min Spacing = 3λ.

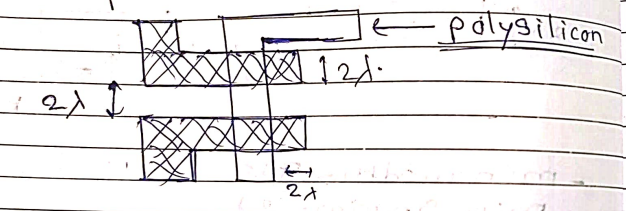
④ Rule for metal 2



- min width = 3λ
- min Spacing = 4λ

⑤ Rule for Poly-1

- min width = 2λ
- min Spacing = 2λ
- min gate extenⁿ = 2λ



⑥ For Via-1

- Exact Size = 2λ
- Min Spacing = 1λ
- Min Spacing = 2λ

⑦ for Metal-3

- Minimum Size = λ
- Minimum Spacing = 4λ

* Spice code CMOS Inverter AC-analysis

→

```
VDD Vdd 0 5        ; 5V Supply
Vin In 0 AC 1      ; Ac input source
M1 out In 0 0 NMOS ; [Pull-down]
M2 out In Vdd Vdd PMOS; (Pull up)
C1 out 0 10p       ; load capacitor
.ac dec 10 1 meg   ; Ac sweep from 1Hz to 1 MHz
.model NMOS NMOS   ; NMOS model defin.
.model PMOS PMOS   ; PMOS model defn.
.end               ; end of simuln.
```

* Need design rules in VLSI

1. To ensure correct fabrication
   - manufactured without errors
2. To avoid physical defects
   - prevent issues like SC, OC.
3. To maintain minimum Spacing and width.

4. To reduce cost and manufacturing Complexity.

5. To ensures peeformance and reliability.

* Design Rule check

① Design Rule check is an automatic process.
- DRC Verifies a chip physicad layout. The layout must Specifying rules.
- These rules aree called design rules.

- The main purpose is to ensure the design is manufactuacable.
- It Peevent design issue.

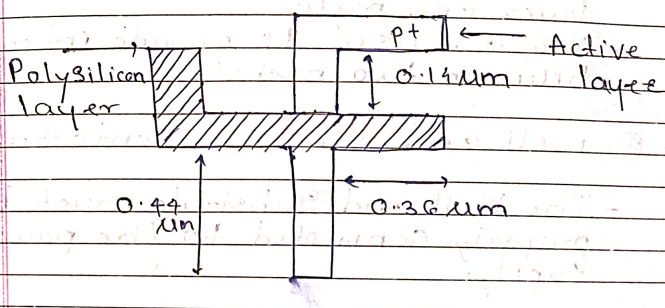
① Antenna Rule → check the metal line that are too long beforee connectly to a Protedive diode.

layout design Rule.

① Transistor rules.
② Contact X via rules.

③ well and Substrate Contact rule.

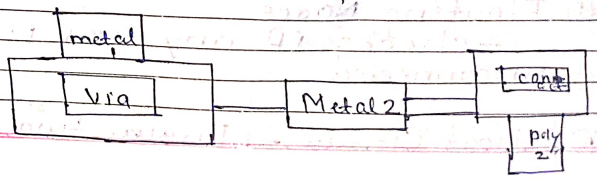
① Transistor rule.



- min length of a Transistor = min width of Poly Silicon = 24 um.
- width of the Transistor = min width of the active layee = 0.3 um

② Contact and via types

- These rules aree explained in foll



- The Contact and via are used for interconnections.
- The Contacts are used to make an interconnection between polysilicon layer and metal
- Via are used to provide interconn between two metal.

③ well and Substrate Contact rule.

- The Well and Substrate must be properly connected to the power supply.

* Electric Rule check

① Electrical Rule is a Vertification step in VLSI design.
② It is often performed after physical layout is Complete.
③ The main goal of ERC is to ensure Circuit electrically Correct.

It checks

① Floating Nodes
- checks if any pins is left unconnected

② Short Circuit → Ensure two different nets are not wrongly shorted.

3. Power and ground Connection mistakes

4. wrong pin connection.

* Process → The layout is extracted to Create a netlist.
The ERC Tool Analyzes netlist.
It generates a report of all ERC Violations.

Steps involved in fabrication of CMOS transistor

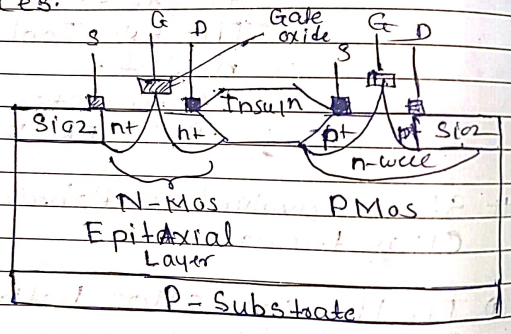
① In the Case CMOS, both n-mos & P-MOS exit.

- The N-MOS and P-MOS are fabricated on the same chip to form a CMOS Technology.

- NMOS → P-Type & PMOS → N-Type.

- To place both transistor on the one chip, Special Technology is used Called WELL Technology.

- N-MOS is made directly on P-Type Substrate.
- P-MOS is made inside N-well, which is specially created in the p-substrate.
- SiO_2 layer provide isolation b/w devices.



- The Terminal Sources Gate, Drain are connected using (n+) regions for n-MOS.

The CMOS fabrication steps.

Define active areas
↓
Etch & fill Trenches
↓
Implant well regions
↓
Deposit polysilicon layer
↓
Implant Source & Drain
↓
Deposit Contact

Antenna effect

- During CMOS fabaication Plasma is used for etching and deposition.
- when long metal wire connected to the gate terminal of MOSFET. & to collect electric charge.
- These wires behave like an antenna

So this problem is called Antenna Effect.

- So interconnected wire to the gate terminal acts as an antenna.
- There is a diode that is formed between Drain and Sources diffusion layers.
- The diode starts conducting.

Effect of Antenna Effect

- Threshold level gets shifted.
- Reduced Carrier mobility.
- The worst case, GATE failure
- Poor VI characteristic.

Reduce

① Control and optimize Plasma processing steps to reduced charges building.

② Use proper fabrication method to avoid excesive charge accumulation

③ use jumpers instead using long wire.

Cross-talk

- Crosstalk is an unwanted interference, that happens when one single wire affects another nearby wire in VLSI Circuit.
- It occues because wire are placed Very close together and the electric Signal from one wire induced noise on another wire.
- happen to parasitic Capacitance & indunctance.
- wires placed very close in high-density.

Effects

- Wrong Vtg Spike causes wrong o/p.
- Signal become slower or faster. affecting Timing.
- Extra Switching increases power Consumption.

Reduce Crosstalk

- Increase distance bet^n Parallel wire.
- place ground or power lines between

wires.
- Use buffer to improve Signal strength.

* Electro Migration

- Electromigration is the movement of metal atoms in a wire due to high current flow.

- When current density is high, fast moving electrons push the atoms causing to the wire

Effect

- Void / open ckt
  - Atom move away and Created gaps, breaking the wire.
- Hillock / SC ckt :
  - Atom pile up and may touch nearby wires.

Causes
- High Temperature.
- High current density.
- material Structure.

Steps to reduce
- Increase metal width.
- Reduce the power Supply.
- Reduce the Supply frequency.
- Use short wires.

* Stick diagram

watch from barapate tutorials.
yt